VINCENZO CECCARELLI GRIMALDI

Research Assistant/Associate (f/m/d) – Doctoral Candidate, ERC Synergy Grant SAFEr Grid (V000010837)

CONTACT

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STRENGTHS

Energy Systems Modeling
Physics-informed Digital Twins, Cyber-Physical Energy Systems Modeling & Simulation, Smart Grid Stability Analysis, Asynchronous Subgrid Operation, DER Integration, Predictive Modeling, Real-Time Co-Simulation

Intelligent Control
Reinforcement Learning for Adaptive Grid Control, Multi-Agent Systems, Model Predictive Control (MPC), Agent-Based Stability Optimization, Python (Tensorflow, PyTorch, Pandas, NumPy, Scikit-learn), Matlab/Simulink, Git, Linux, Docker

INTERESTS

- Cyber-Physical Power Systems
- Physics-Informed Digital Twins
- Smart Grid Stability & Dynamics
- Asynchronous Operation of Modular Subgrids
- Agent-Based Control & Coordination
- Decentralized Energy Resource Management
- Real-Time Simulation & Adaptive Control
- Store-and-Forward Energy Architectures

PROFESSIONAL SUMMARY

As an Electrical Business Engineer with a specialization in power engineering, I have developed a strong foundation in the modeling, automation, and control of electrical power systems. My academic and project work has focused on smart grids, physics-informed digital twins, reinforcement learning, and decentralized energy system architectures. I am particularly interested in how advanced control, interoperable digital infrastructures, and modular subgrid concepts can enable the stable and reliable operation of future low-inertia, asynchronous power systems. Technical depth and systems thinking come together to create innovative control strategies for resilient, decentralized power grids.

RESEARCH-ORIENTED & DIGITAL WORK SAMPLES

- **GridOS**: Self-initiated modular software project for decentralized energy systems, including components for multi-protocol connectivity (IEC 61850, DNP3, Modbus, MQTT, OPC UA), digital-twin-oriented modelling, forecasting, and optimization; developed to investigate data flows, control architectures, and coordination mechanisms in DER-based and modular grid environments.
- **NeuralBridge**: Middleware project with modular adapter and API structures for cross-domain data exchange, as well as modules for authentication, role-based access control, auditing, and compliance-related documentation; used as a practical framework for exploring interoperability, secure communication, and governance in complex cyber-physical systems.
- **Methodological Focus**: Application of learning-based models for the analysis, modelling and digital energy infrastructures.

EDUCATION

M.Sc. Business Administration & Engineering: Electrical Power Engineering

Overall Grade: 2.7

Aug 2022 – Aug 2025 RWTH Aachen University, NRW

- **Master's thesis (Grade 2.3):** Data Modeling in a Cross-domain Ontology for Cyber Intelligence in Smart Grids Using Reinforcement Learning

B.Sc. Business Administration & Engineering: Electrical Power Engineering

Overall Grade: 3.5

Sept 2018 – Aug 2022 RWTH Aachen University, NRW

- **Bachelor's thesis (Grade 3.0):** Development of a Cross-domain Knowledge Base for Cyber Intelligence in Smart Grids

ACCREDITATION

ITK Fachspezialist

DB InfraGO AG

01 Jan 2026 Berlin

LANGUAGES

- German — Professional
- English — Professional
- Spanish — Professional
- Chinese Mandarin — Professional

RELEVANT WORK EXPERIENCE

ITK Specialist

DB InfraGO AG - Technische Experten / Wissensmanagement 1 (V.IO-MI-IW 1)

Aug 2024 – Present Frankfurt am Main, HE

- Developed learning-based adaptive control strategies for stability analysis in electrified railway power systems under highly variable loads, using TensorFlow and PyTorch.
- Designed interoperable communication and control architectures for safety-critical cyber-physical railway power systems.
- Conduct technical plan reviews, commissioning support, and system validation to help ensure the reliable operation of high- and medium-voltage power systems.

Industrial Internships (Voluntary Work Placement)

DB Fahrzeuginstandhaltung GmbH - Vertriebliches Angebotsmanagement und Vertragsadministration (P.WVV 34)

Dec 2022 – May 2024 Frankfurt am Main, HE

- Performed dielectric testing and ultrasonic inspection of high-voltage components to assess insulation integrity and flashover-related risks in operational systems.
- Supported reliability-oriented analysis of electrically relevant components and contributed to technically precise maintenance and condition-assessment processes.

Industrial Internship (for the Bachelor Degree)

DB Netz AG - Supplier and System Testing IH (V.IWM 17)

Aug 2022 – Dec 2022 Frankfurt am Main, GER

- Supported quality assurance and functional testing of communication components in safety-critical signalling systems.
- Performed time-series analysis of operational data to support the assessment of system reliability and operational stability.